



# Modelling Financial Markets

## Seminar 15: LLMs and Sentiment Analysis — Hands-on



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## Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AUC: Area Under the (ROC) Curve
- ▣ BERT: Bidirectional Encoder Representations from Transformers
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAR: Cumulative Abnormal Return
- ▣ CI: Confidence Interval
- ▣ ECE: Expected Calibration Error
- ▣ FNSPID: Financial News and Stock Price Integration Dataset
- ▣ GPT-4: Generative Pre-trained Transformer 4
- ▣ LLM: Large Language Model
- ▣ LM: Loughran-McDonald (financial sentiment dictionary)
- ▣ NLL: Negative Log-Likelihood
- ▣ OLS: Ordinary Least Squares
- ▣ PPI: Prediction-Powered Inference
- ▣ SD: Standard Deviation
- ▣ SE: Standard Error
- ▣ TF-IDF: Term Frequency - Inverse Document Frequency
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)

## Today's Questions and Route

- ▣ Questions of the seminar
  - ▶ can a language model read financial sentiment?
  - ▶ does that sentiment predict returns one can still trade?
- ▣ Route
  - ▶ First the primer “What You Need for Today” (three slides): [Chapter 15](#) comes after the seminar
  - ▶ A1, A3, A7 on paper: how a misclassified label attenuates a slope, how precise a paired accuracy test is, at what cost a strategy stops earning
  - ▶ B1: is FinBERT better than a dictionary?
  - ▶ B5: do headlines predict returns or describe them?
  - ▶ B7: does the LLM remember the market?
  - ▶ C2: find the three errors in an AI answer; B2 at home
- ▣ What you produce: the A answers, one table and one figure per B exercise, the C2 corrections, one interpretation sentence per exercise
- ▣ Notebook for today: [open the seminar notebook in Colab](#); every exercise has a section with the same label
- ▣ Seminar: Antoaneta Amza, PhD candidate, [antoaneta.amza@csie.ase.ro](mailto:antoaneta.amza@csie.ase.ro)

## Exercise Map

| Exercise        | Question                                                                        | Type     | Model                 |
|-----------------|---------------------------------------------------------------------------------|----------|-----------------------|
| A1              | how much does a misclassified dummy shrink a slope?                             | Solved   | –                     |
| A3              | how precise, and how powerful, is a paired accuracy test?                       | Solved   | –                     |
| A7              | at what cost per trade does the strategy stop earning?                          | Solved   | –                     |
| B1              | is FinBERT more accurate than the LM dictionary?                                | Solved   | –                     |
| B5              | do headlines predict returns, or describe them?                                 | Solved   | –                     |
| B7              | does the LLM remember the market?                                               | Solved   | –                     |
| C2              | what is wrong in an AI answer?                                                  | Proposed | B1, B5, B7            |
| B2              | is a 7-billion-parameter LLM better than FinBERT?<br>(homework)                 | Proposed | B1                    |
| Extension packs | A2, A4, A5, A6, A8, B3, B4, B6, B8–B11, C1, C3 (optional,<br>after the lecture) | both     | named on each<br>task |

- ▣ **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- ▣ Part A: derivations on paper; Part B: verification and inference on real data, each ending with an interpretation question; Part C: open questions and AI critique

## Data Used

| Data                                                      | Content                                                                                                | Saved scores                         | Used in                         |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------|---------------------------------|
| Twitter Financial News (validation set, Hugging Face Hub) | 2 388 headlines from 2022: 1 566 neutral, 475 positive, 347 negative                                   | LM, FinBERT, Qwen2.5 (P1–P3), MiniLM | A1–A5, A8, B1–B4, B8, B9, C2    |
| Financial PhraseBank <a href="#">Malo et al. (2014)</a>   | 4 846 news sentences, 2 264 with full agreement                                                        | same scores                          | B3                              |
| FNSPID <a href="#">Dong, Fan &amp; Peng (2024)</a>        | 106 820 headlines on 16 US stocks, 27.10.2009–18.12.2023                                               | daily LM, FinBERT, Qwen2.5-7B scores | A7, A8, B5, B6, B10, B11, C1–C3 |
| Stocks and SPY                                            | adjusted daily prices of the course data, one series per asset (field used: adjusted close), 2009–2023 | –                                    | B5, B6, C1, C3                  |
| S&P 500                                                   | monthly closes, Jan 2000 – Aug 2026                                                                    | Qwen2.5 answers about each month     | B7, C2                          |

- ▣ The scores are saved with the chapter's Quantlets, [Quantlets/Ch\\_15](#): no language model has to run in class
- ▣ C3 (extension pack) needs a new scoring run of the FNSPID headlines

## Setup: Run and Verify

```
# notebook section "Setup" defines the functions that
# read the data
pb, tw = eval_texts()    # PhraseBank, Twitter
P, rets = news_panel()   # FNSPID stock-days
print(len(pb), len(tw))
print(tw['label'].value_counts())
print(P.shape,
      P.index.get_level_values('ticker').nunique())
```

- Expected output
  - ▶ 4 846 and 2 388 rows
  - ▶ neutral 1 566, positive 475, negative 347
  - ▶ (18 282, 19) and 16 stocks
- Each row of tw holds the text, the human label and the probabilities of every model (e.g. finbert\_negative, qwen7B\_positive)
- If your counts differ, stop and check the files before any exercise

## Notation Used in Every Exercise

- Dictionary tone  $\tau = (P - N)/(P + N)$ , with  $P$ ,  $N$  the counts of positive and negative words; LM: Loughran–McDonald financial lists [Loughran & McDonald \(2011\)](#)
- Model probabilities  $p_k = e^{z_k/T} / \sum_j e^{z_j/T}$  ( $z_k$ : logit of answer  $k$ ,  $T$ : temperature); label  $\arg \max_k p_k$ ; net score  $s = p_{\text{pos}} - p_{\text{neg}}$ ; FinBERT [Araci \(2019\)](#), Qwen2.5 [Qwen Team \(2024\)](#)
- Two classifiers A and B on the same texts:  $n_{01}$  = A wrong and B right,  $n_{10}$  = A right and B wrong; McNemar test [McNemar \(1947\)](#): under  $H_0$ ,  $n_{01} \sim \text{Bin}(n_{01} + n_{10}, 0.5)$
- Day  $d$ : the first trading day on or after the headline date;  $d + 2$ : the first return surely after publication;  $h$ : days after  $d$
- bp: basis point, 0.01%; SE: standard error; CI: 95% confidence interval;  $t_{NW}$ : Newey–West  $t$  [Newey & West \(1987\)](#); clustered SE: [Petersen \(2009\)](#)

## What You Need for Today (1/3): Three Ways to Read Sentiment

- **Chapter 15** comes after this seminar: these three slides are the minimum for today's core tasks; depth and evidence in the lecture
- **Dictionary** (bag of words): count the positive ( $P$ ) and negative ( $N$ ) words of fixed lists; LM lists were built from 10-K filings [Loughran & McDonald \(2011\)](#), with 2355 negative and only 354 positive words
  - ▶ Tone  $\tau = (P - N)/(P + N)$ , class = sign of  $\tau$ ; example  $P = 2$ ,  $N = 1$ :  $\tau = 1/3$ , positive; transparent, but blind to context and negation
- **FinBERT** [Araci \(2019\)](#): BERT, a Transformer encoder ([Chapter 14](#)), further trained on financial news and fine-tuned on labelled sentences; output:  $p_{\text{pos}}$ ,  $p_{\text{neu}}$ ,  $p_{\text{neg}}$ 
  - ▶ The public model was fine-tuned on Financial PhraseBank [Malo et al. \(2014\)](#): accuracy there is partly in-sample
- **LLM zero-shot** (Qwen2.5 [Qwen Team \(2024\)](#)): a large language model asked "Is this headline positive, negative or neutral?" with no training on these labels; we read the logits  $z_k$  of the three answer words
  - ▶ Softmax with temperature  $T$ :  $p_k = e^{z_k/T} / \sum_j e^{z_j/T}$ ; net score  $s = p_{\text{pos}} - p_{\text{neg}} \in [-1, 1]$
  - ▶ Example  $z = (2, 1, 0)$  for (pos, neu, neg):  $T = 1$  gives  $p = (0.665, 0.245, 0.090)$ ,  $s = 0.575$ ;  $T = 2$  flattens to  $(0.506, 0.307, 0.186)$ ,  $s = 0.320$

## What You Need for Today (2/3): Scoring a Sentiment Classifier

- Confusion matrix: rows = human labels (the truth), columns = predicted labels; accuracy = diagonal / total
- Example: 100 headlines, 20 truly negative; the model flags 18 of them and 8 of the other 80
  - ▶ Rates:  $\pi = 0.20$  truly negative,  $p = 0.26$  flagged, misses  $\alpha_1 = 2/20 = 0.10$ , false alarms  $\alpha_0 = 8/80 = 0.10$  (the inputs of A1)
  - ▶ Accuracy  $(18 + 72)/100 = 0.90$ , but “never negative” already scores 0.80: compare with the majority class, never with  $1/K$
  - ▶ Macro-F1: the mean over classes of  $F_1 = 2 \text{ prec} \cdot \text{rec} / (\text{prec} + \text{rec})$ ; here  $(0.783 + 0.935)/2 = 0.86$ , while “never negative” gets  $(0 + 0.889)/2 = 0.44$
- Two classifiers on the same headlines are **paired**: only the discordant headlines ( $n_{01}$ ,  $n_{10}$ ) carry information about which is better (McNemar test, notation slide; A3, B1)
- A misclassified label is a regressor measured with error: a slope on it is biased towards zero (attenuation, [Aigner \(1973\)](#); derived in A1)

## What You Need for Today (3/3): From Headlines to Returns

- Timing: a headline dated on a calendar day maps to day  $d$ , the first trading day on or after that date; the publication hour is unknown, so the return of  $d + 1$  may still contain the reaction
  - ▶ Example: headline on Saturday  $\Rightarrow d = \text{Monday}$ ; buy at the close of Tuesday ( $d + 1$ ); the first return one can trade is Wednesday's ( $d + 2$ )
  - ▶ A slope at  $h = 0$  measures how headlines *describe* the day; only  $h \geq 2$  is a *forecast* (B5)
- Panel regression  $r_{i,d+h}^e = a + b z_{i,d} + \varepsilon$  on stock-days; errors are correlated within a day (common news) and within a stock (over time)
  - ▶ Clustered SE [Petersen \(2009\)](#):  $\widehat{\text{Var}}(\hat{b}) = (X^\top X)^{-1} (\sum_g X_g^\top \hat{u}_g \hat{u}_g^\top X_g) (X^\top X)^{-1}$ , sum over clusters  $g$  (days or stocks)
  - ▶ Only  $G$  clusters: with 16 stocks use  $t_{G-1}$  critical values (2.13) or a wild cluster bootstrap [Cameron, Gelbach & Miller \(2008\)](#)
- Sentiment strategy: at the close of  $d + 1$ , a position of sign  $s_{i,d}$  in each stock with news, hedged with SPY, closed one day later; cost  $c$  per trade, four trades on each day with positions (A7)
- Look-ahead bias [Glasserman & Lin \(2023\)](#): an LLM trained on text up to its release may have read what followed a headline; a clean test uses only data after the weights were published (B7, C2)
- More in [Chapter 15](#), after the seminar: embeddings, calibration, the scores as estimates and the ChatGPT case study

## A1 [Solved]: Attenuation from Misclassification — Task

- **Question:** a regression uses the FinBERT label "negative" instead of the truth; by how much is the slope shrunk?
- **Data**
  - ▶ Twitter Financial News, validation set:  $N = 2\,388$  headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
  - ▶ Model  $r = a + \beta s^* + \varepsilon$ ,  $s^* = 1$  if the headline is truly negative; we observe only  $\hat{s} = 1$  when FinBERT predicts negative
  - ▶ Non-differential error: given  $s^*$ ,  $\hat{s}$  says nothing about  $r$ , so  $\text{Cov}(\varepsilon, \hat{s}) = 0$
  - ▶  $\pi = P(s^* = 1)$ ,  $p = P(\hat{s} = 1)$ , false alarms  
 $\alpha_0 = P(\hat{s} = 1 \mid s^* = 0)$ , misses  $\alpha_1 = P(\hat{s} = 0 \mid s^* = 1)$
- **Tasks**
  - (a) Show that  $\text{plim } \hat{b} = \beta \text{Cov}(s^*, \hat{s}) / \text{Var}(\hat{s})$ .
  - (b) Derive  $\lambda = \pi(1 - \pi)(1 - \alpha_0 - \alpha_1) / [p(1 - p)]$ .
  - (c) Compute  $\pi$ ,  $p$ ,  $\alpha_0$ ,  $\alpha_1$  from the matrix, then  $\lambda$  to three decimals.
  - (d) Compute  $\text{Cov}(u, s^*)$  for  $u = \hat{s} - s^*$ : is the error classical?
- **Report:**  $\lambda$ ,  $1/\lambda$  and the sign of  $\text{Cov}(u, s^*)$
- **Interpretation:** what share of a true effect survives in a slope on the FinBERT dummy?

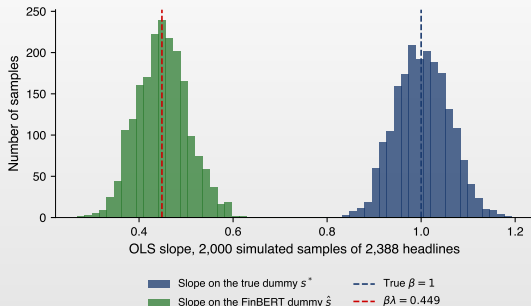
FinBERT confusion matrix, Twitter

| True \ predicted | neg | neu   | pos | Total |
|------------------|-----|-------|-----|-------|
| neg              | 264 | 71    | 12  | 347   |
| neu              | 213 | 1 184 | 169 | 1 566 |
| pos              | 58  | 133   | 284 | 475   |
| Total            | 535 | 1 388 | 465 | 2 388 |

## A1 [Solved]: Solution (1/2)

- ▣ (a) For one regressor  $\text{plim } \hat{b} = \text{Cov}(r, \hat{s}) / \text{Var}(\hat{s})$  and  $\text{Cov}(r, \hat{s}) = \beta \text{Cov}(s^*, \hat{s}) + \text{Cov}(\varepsilon, \hat{s})$ ; the last term is 0 by the non-differential assumption
- ▣ (b) Both variables are binary:  $\text{Cov}(s^*, \hat{s}) = P(s^* = 1, \hat{s} = 1) - \pi p = \pi(1 - \alpha_1) - \pi p$
- ▣ Total probability:  $p = \pi(1 - \alpha_1) + (1 - \pi)\alpha_0$ ; substituting,  $\text{Cov}(s^*, \hat{s}) = \pi(1 - \pi)(1 - \alpha_0 - \alpha_1)$
- ▣  $\hat{s}$  is Bernoulli, so  $\text{Var}(\hat{s}) = p(1 - p)$ ; the ratio of the two is  $\lambda$

## A1 [Solved]: Solution (2/2)



- (c)  $\pi = 347/2\,388 = 0.1453$ ,  $p = 535/2\,388 = 0.2240$ ,  $\alpha_1 = 1 - 264/347 = 0.2392$ ,  $\alpha_0 = 271/2\,041 = 0.1328$
- $\lambda = \frac{0.1242 \times 0.6280}{0.2240 \times (1 - 0.2240)} = 0.449$ , so  $\beta = 2.23 \hat{b}$
- (d)  $\text{Cov}(u, s^*) = 0.0780 - 0.1242 = -0.0462 < 0$ : the error is negatively correlated with the truth, not classical [Aigner \(1973\)](#)
- Chart: 2 000 simulated samples with these rates and  $\beta = 1$ ; the slope on  $\hat{s}$  averages 0.450 (SD 0.054), as  $\lambda$  predicts
- Interpretation: a slope on the FinBERT dummy keeps only about 45% of the true effect

## A3 [Solved]: Paired Accuracy Difference: Variance, Exact Test, Power — Task

### ▣ Questions

- ▶ how precise is the accuracy gap between two classifiers scored on the same headlines?
- ▶ how many headlines does it need?

### ▣ Inputs

- ▶ Twitter validation set,  $n = 2\,388$  independent headlines; A = LM dictionary, B = FinBERT
- ▶ Discordant counts:  $n_{01} = 539$  (A wrong, B right),  $n_{10} = 252$  (A right, B wrong)
- ▶  $D_i = 1$  if A wrong and B right,  $-1$  if A right and B wrong,  $0$  otherwise;  $d = ED_i = \text{acc}_B - \text{acc}_A$ ,  $q = P(D_i \neq 0)$

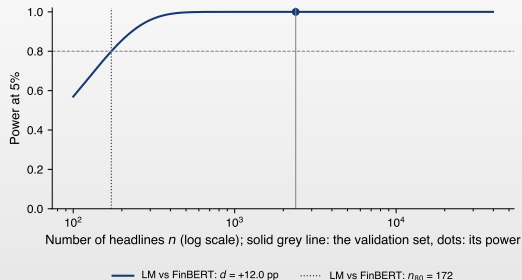
### ▣ Tasks

- (a) Show  $\hat{d} = (n_{01} - n_{10})/n$  and derive  $\text{Var}(\hat{d}) = (q - d^2)/n$ .
- (b) Compute  $\hat{d}$ , SE, the 95% CI and  $z = \hat{d}/\text{SE}$ .
- (c) Run the exact McNemar test at 5% and state the decision.
- (d) Compute the power  $\Phi(|d|\sqrt{n}/\sqrt{q - d^2} - 1.96)$  if the true  $d$  is  $\hat{d}$ , and  $n_{80} = (1.96 + 0.84)^2(q - d^2)/d^2$ .

- ▣ **Report:**  $\hat{d}$ , SE, CI,  $z$ , exact  $p$ , the decision, the power and  $n_{80}$

- ▣ **Interpretation:** which headlines determine the variance of  $\hat{d}$ ?

## A3 [Solved]: Solution



- (a)  $\hat{d} = \frac{1}{n} \sum_i D_i = (n_{01} - n_{10})/n$ ;  $D_i^2 = 1$  exactly when one is right and the other wrong, so  $ED_i^2 = q$ ,  $\text{Var}(D_i) = q - d^2$  and  $\text{Var}(\hat{d}) = (q - d^2)/n$
- (b)  $\hat{d} = (539 - 252)/2388 = 0.1202$ ,  
 $\hat{q} = 791/2388 = 0.3312$ ,  $\text{SE} = 1.15$  pp,  $\text{CI} [9.76; 14.28]$  pp,  $z = 10.43$
- (c) Under  $H_0$  we expect 395.5 on each side; exact  $p < 0.001$ : reject equal accuracy
- (d) Power  $\Phi(+8.47) \approx 100\%$ ;  
 $n_{80} = 7.84 \times (0.3312 - 0.01444)/0.01444 \approx 170$  headlines (chart)
- Interpretation: only the discordant headlines ( $q$ ) enter the variance; a 12-point gap is detectable with a few hundred headlines

## A7 [Solved]: Break-Even Cost with a Standard Error — Task

### ▣ Questions

- ▶ at what cost per trade does the FinBERT strategy stop earning?
- ▶ how precisely do we know that cost?

### ▣ Inputs

- ▶ Sentiment strategy (primer slide 3/3; studied in [Chapter 15](#)): at the close of  $d + 1$  take  $\text{sign}(s_{i,d})$  in every stock with news on day  $d$ , hedged with SPY, closed one day later; four trades on each day with positions; 3568 trading days
- ▶ FinBERT signal: mean gross return over all days  $\hat{\mu} = 3.27$  bp,  $t_{NW} = 1.27$  ( $L = 8$  lags), share of days with positions  $a = 0.962$ , gross Sharpe ratio  $\sqrt{252} \hat{\mu} / \sigma = 0.36$

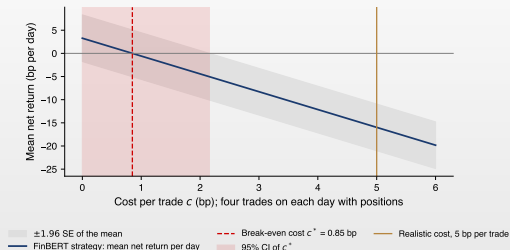
### ▣ Tasks

- Write the mean net daily return and the net Sharpe ratio as functions of the cost  $c$  per trade.
- Derive the break-even cost  $c^*$  at which the mean net return is zero.
- Compute  $\text{SE}(\hat{\mu}) = \hat{\mu} / t$ , then  $\text{SE}(c^*)$  by the delta method and the 95% CI of  $c^*$ .
- Recover  $\sigma$  from the gross Sharpe ratio and compute the net Sharpe ratio at  $c = 5$  bp.

▣ **Report:**  $c^*$ , its SE and CI, the net Sharpe ratio

▣ **Interpretation:** is a realistic cost of 5 bp per trade inside the CI of  $c^*$ ?

## A7 [Solved]: Solution



- (a) On a share  $a$  of days we pay  $4c$ :  $\mu_{\text{net}} = \mu - 4ac$ ,  
 $SR_{\text{net}} = \sqrt{252} (\mu - 4ac) / \sigma$ , with  $\sigma$  barely changed by costs
- (b)  $c^* = \mu / (4a) = 3.27 / 3.85 = 0.85$  bp per trade
- (c)  $SE(\hat{\mu}) = 3.27 / 1.27 = 2.57$  bp; with  $a$  fixed by the news calendar,  $SE(c^*) = 2.57 / 3.85 = 0.67$  bp, CI  $[-0.46; 2.16]$  bp
- (d)  $\sigma = \sqrt{252} \times 3.27 / 0.36 = 143$  bp;  
 $SR_{\text{net}} = \sqrt{252} (3.27 - 3.85 \times 5) / 143 = -1.77$
- Interpretation: 5 bp lies far above the CI of  $c^*$ ; with daily turnover the strategy cannot survive realistic costs

 MFM\_ch15\_sem\_partA

## B1 [Solved]: Is FinBERT Better than LM? — Task

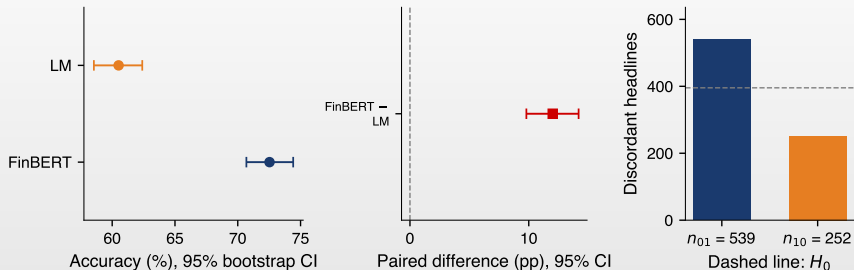
- **Question:** is the fine-tuned FinBERT more accurate than the Loughran–McDonald dictionary on headlines it was not trained on?
- **Data**
  - ▶ Twitter Financial News, validation set:  $N = 2\,388$  headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
  - ▶  $A$  = LM dictionary (label = sign of the tone),  $B$  = FinBERT (label = largest probability); notebook function `b_pairs`
- **Tasks**
  - (a) Compute each accuracy with a percentile bootstrap 95% CI: resample the headlines 2 000 times (seed 0).
  - (b) Compute the difference  $B - A$  with a paired bootstrap CI: the same resampled headlines for both classifiers.
  - (c) Count  $n_{01}$ ,  $n_{10}$ ; run the exact McNemar test and  $(n_{01} - n_{10})^2 / (n_{01} + n_{10})$  against  $\chi^2_1$  (3.84).
- **Report:** a table of the two accuracies, their CIs and the difference;  $n_{01}$ ,  $n_{10}$ , both  $p$ -values and the decision
- **Interpretation**
  - ▶ why is the paired test more powerful than comparing the two intervals?
  - ▶ does the gap matter for trading?

## B1 [Solved]: Solution (1/2)

|                 | Accuracy (%) | CI 95%        |
|-----------------|--------------|---------------|
| LM              | 60.5         | [58.5; 62.4]  |
| FinBERT         | 72.5         | [70.7; 74.4]  |
| Difference (pp) | +12.0        | [+9.8; +14.2] |

- Step 1: labels of both classifiers on the 2 388 headlines; step 2: 2 000 bootstrap resamples of the headline indices; step 3: both accuracies and their difference in each resample
- (a)–(b) The two intervals do not overlap and the paired CI of the difference excludes 0; it agrees with the analytic CI of A3

## B1 [Solved]: Solution (2/2)



- (c)  $n_{01} = 539$ ,  $n_{10} = 252$ , 395.5 expected under  $H_0$ ; exact  $p < 0.001$ ;  $\chi^2 = 104.1 > 3.84$ : reject equal accuracy
- Pairing removes the variation in headline difficulty: headlines both get right, or both get wrong, carry no information
- Interpretation: FinBERT is clearly better than the dictionary outside its training data; a 12-point gain in labelling says nothing yet about returns (B5)

## B5 [Solved]: Reaction or Prediction? — Task

### □ Questions

- ▶ does the daily headline score explain the return of the news day?
- ▶ does it forecast the first return one can still trade?
- ▶ how wide are the honest intervals?

### □ Data

- ▶ FNSPID panel: 18 282 stock-days with news, 16 US stocks, 27.10.2009–18.12.2023;  $z_{i,d}$ : daily net score, standardised over the panel;  $r^e$ : excess return over SPY (%), from the adjusted daily closes of the course data
- ▶ Scores: FinBERT, Qwen2.5-7B, LM

### □ Tasks

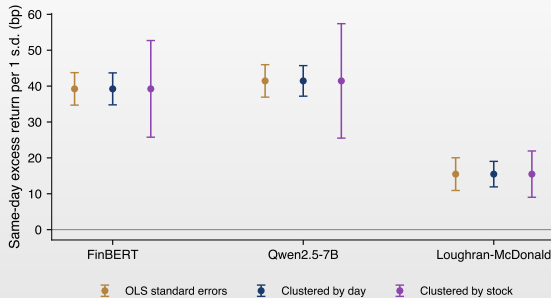
- (a) Estimate  $r_{i,d+h}^e = a + b z_{i,d} + \varepsilon$  by OLS for the three scores and  $h = 0, 1, 2$ ; report  $b$  in bp per one SD of the score.
- (b) Compute three SEs: OLS, clustered by day ( $G = 3\,434$ ), clustered by stock ( $G = 16$ ), with the small-sample factor  $\frac{G}{G-1} \frac{n-1}{n-2}$ .
- (c) Compare each  $t$  with 1.96, and the stock-clustered  $t$  with the  $t_{15}$  critical value 2.13.

### □ Report: a chart of the $h = 0$ slopes with the three intervals, and a table of the regressions

### □ Interpretation

- ▶ why do  $h = 0$  and  $h = 2$  differ?
- ▶ can a clustered SE with 16 clusters be trusted?

## B5 [Solved]: Solution (1/2)



- Same-day slopes with the 95% intervals of the three standard errors: the stock-clustered interval is the widest
- Clustering by day is close to OLS: with only 5.3 stocks per day there is little common noise
- Clustering by stock multiplies the FinBERT SE at  $h = 0$  by 3.0: the slope differs across stocks and each stock's errors are correlated over time

## B5 [Solved]: Solution (2/2)

| Score      | $h$ | $b$ (bp) | SE OLS | SE day | SE stock | $t$ day | $t$ stock |
|------------|-----|----------|--------|--------|----------|---------|-----------|
| FinBERT    | 0   | +39.2    | 2.3    | 2.3    | 6.9      | +17.27  | +5.71     |
| FinBERT    | 2   | -0.5     | 2.2    | 2.2    | 1.5      | -0.25   | -0.35     |
| Qwen2.5-7B | 0   | +41.4    | 2.3    | 2.2    | 8.1      | +19.07  | +5.10     |
| Qwen2.5-7B | 2   | +2.3     | 2.2    | 1.9    | 1.0      | +1.19   | +2.41     |
| LM         | 0   | +15.5    | 2.3    | 1.8    | 3.3      | +8.53   | +4.70     |
| LM         | 2   | -0.0     | 2.2    | 1.7    | 0.9      | -0.01   | -0.03     |

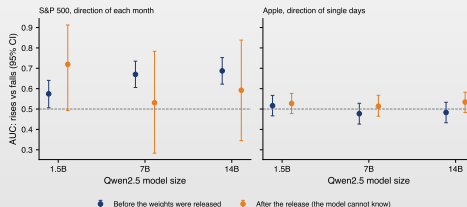
- $h = 0$ : large under every SE ( $t$  by stock above 4.7): headlines describe the day's move, often the move itself ("shares jump")
- $h = 2$ : FinBERT -0.5 bp ( $t = -0.25$  by day); Qwen  $t = +2.41$  by stock, but  $p = 0.061$  by wild cluster bootstrap [Cameron, Gelbach & Miller \(2008\)](#) and 0.37 after Holm over nine regressions (extension B10)
- Interpretation: sentiment explains the news day, not the tradable days; with 16 stocks, inference needs few-cluster methods

## B7 [Solved]: Does the Model Remember the Market? — Task

- **Question:** asked without any data, can Qwen2.5 tell the months in which the S&P 500 rose from those in which it fell?
- **Data**
  - ▶ S&P 500 monthly closes (GSPC.INDX), January 2000 to August 2026; a month is "up" if its return is positive
  - ▶ Qwen2.5 1.5B, 7B, 14B (weights published 19 September 2024) asked "Did the S&P 500 index rise or fall during ⟨month year⟩?"; score  $P(\text{rise})$
  - ▶ Before: 296 months to August 2024 (185 up); after: 23 months from October 2024 (13 up)
- **Tasks**
  - (a) Compute the  $\text{AUC} = U / (n_{\text{up}} n_{\text{down}})$  ( $U$ : Mann–Whitney) per model and period, with a 2000-draw bootstrap CI over months.
  - (b) Permutation test for 14B before the release: shuffle the up/down labels 2000 times.
  - (c) Test  $\text{AUC}_{\text{before}} = \text{AUC}_{\text{after}}$  with DeLong variances [DeLong et al. \(1988\)](#) and find the smallest drop detectable with 80% power.
- **Report:** the AUC table with CIs, the permutation  $p$ , the DeLong  $z$  and  $p$ , the detectable drop
- **Interpretation:** what does an AUC above 0.5 mean when the model sees no data?

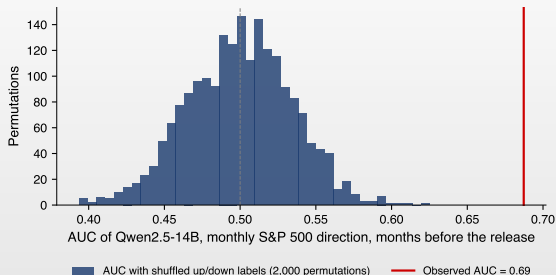
## B7 [Solved]: Solution (1/2)

| Qwen2.5 | AUC before | CI 95%       | AUC after | CI 95%       |
|---------|------------|--------------|-----------|--------------|
| 1.5B    | 0.57       | [0.51; 0.64] | 0.72      | [0.49; 0.91] |
| 7B      | 0.67       | [0.61; 0.73] | 0.53      | [0.28; 0.78] |
| 14B     | 0.69       | [0.62; 0.75] | 0.59      | [0.35; 0.84] |



- (a) 7B and 14B rank up months above down months far above chance before the release; 1.5B barely; after the release all intervals contain 0.5

## B7 [Solved]: Solution (2/2)



- (b) Permutation  $p$  for 14B before the release  $< 0.001$ : no shuffled AUC reaches the observed one (chart)
- (c) DeLong: SE before 0.032, after 0.126;  $z = 0.73$ ,  $p = 0.47$ : do not reject equality; detectable drop  
 $2.80 \sqrt{SE_{\text{before}}^2 + SE_0^2} = 0.36$ , about 101 post-release months needed
- Interpretation: without data, the discrimination comes from the training text; a backtest on pre-release news mixes forecasting with remembering [Glasserman & Lin \(2023\)](#)

## C2 [Proposed]: Critique an AI Answer — the Answer

- **Student prompt:** “I have FNSPID headlines for 16 US stocks, 2009–2023, and daily excess returns vs SPY. Check that Qwen2.5-7B reads sentiment, then test whether its headline sentiment forecasts returns out of sample.”
- **Code of the AI assistant** (qwen\_label, qwen\_score: the zero-shot scorer of primer slide 1/3)

```
import pandas as pd, statsmodels.formula.api as smf
from sklearn.metrics import accuracy_score
tw['pred'] = [qwen_label(t) for t in tw['text']] # Twitter validation set
acc = accuracy_score(tw['label'], tw['pred'])
news['day'] = first_trading_day_on_or_after(news['date'])
news['s'] = [qwen_score(t) for t in news['title']] # P(pos) - P(neg)
d = news.groupby(['ticker', 'day'])['s'].mean().reset_index()
d['z'] = (d['s'] - d['s'].mean()) / d['s'].std()
d = d.merge(rets, on=['ticker', 'day']) # rets: ticker, day, xret (%)
fit = smf.ols('xret ~ z', data=d).fit(cov_type='cluster',
                                     cov_kws={'groups': pd.factorize(d['day'])[0]})
```

## C2 [Proposed]: Critique an AI Answer — Task

- ▣ **Model:** B1, B5 and B7 [Solved]
- ▣ **Conclusion of the AI assistant**
  - ▶ “Accuracy 74.2%, more than twice the 33% of a random guess among three classes: Qwen2.5-7B reads financial sentiment very well.”
  - ▶ “Slope +41.4 bp per standard deviation of the score,  $t = +19.07$  (clustered by day): the day’s headlines forecast the stock’s excess return.”
  - ▶ “Qwen2.5 was never trained on these labels or return series, so the 2009–2023 backtest is fully out of sample.”
- ▣ **Question:** the code runs and the numbers are right, yet the answer contains **three** errors typical of sentiment backtests; which?
- ▣ **Tasks**
  - ▶ (a) Find each error: name the line of code or the sentence and explain why it is wrong.
  - ▶ (b) Say how you would detect it without knowing the answer: a class count, a timestamp check, a release date, a memory test.
  - ▶ (c) Write the corrected code or statement, with the numbers it produces.
  - ▶ (d) Log each error in [AI\\_ERRORS.md](#) with the fields: what the AI answered, what was wrong, how you found out, the correct version.

## B2 [Proposed]: Is a 7-Billion-Parameter LLM Better than FinBERT? — Task

- **Model:** B1 [Solved]
- **Question:** does a general open LLM, used zero-shot, beat the fine-tuned FinBERT on headlines neither was trained on?
- **Data**
  - ▶ Twitter Financial News, validation set:  $N = 2\,388$  headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
  - ▶  $A = \text{FinBERT}$ ,  $B = \text{Qwen2.5-7B}$  zero-shot with prompt P1; neither used any Twitter label
- **Tasks**
  - (a) Compute each accuracy with a percentile bootstrap 95% CI: resample the headlines 2000 times (seed 0).
  - (b) Compute the difference  $B - A$  with a paired bootstrap CI: the same resampled headlines for both classifiers.
  - (c) Count  $n_{01}$ ,  $n_{10}$ ; run the exact McNemar test and  $(n_{01} - n_{10})^2 / (n_{01} + n_{10})$  against  $\chi^2_1$  (3.84).
- **Report:** the same table, tests and chart as in B1
- **Interpretation**
  - ▶ do the two intervals overlap?
  - ▶ does that settle the question?
  - ▶ what would a much larger, slower model still bring?

## Team Project Step

- Choose a text source not covered in class (BVB announcements, Romanian financial news, central-bank statements, crypto posts)
- Label at least 200 texts with two annotators and report the agreement; evaluate only on a held-out part
- Log AI use in [AI\\_USE.md](#) and caught errors in [AI\\_ERRORS.md](#); extension packs are optional; C1 and C3 are project starts

## Extension Packs: What Each One Needs

| Pack                     | Exercises               | Prerequisites                             |
|--------------------------|-------------------------|-------------------------------------------|
| 1. Measurement error     | A2, A6                  | A1                                        |
| 2. Comparing classifiers | A4, B3 [Solved], B4, B8 | A3, B1                                    |
| 3. LLM probabilities     | A5 [Solved]             | Notation slide, lecture calibration chart |
| 4. Valid inference       | B9 [Solved], B10, B11   | B5, B7                                    |
| 5. Signals and costs     | A8, B6, C1              | A7, B5                                    |
| 6. Replication lab       | C3                      | C1, the lecture case study                |

- Each exercise has its own notebook section; do them in order within a pack
- Useful for the team project: C1 and C3 are possible project starts

## A2 [Proposed]: Attenuation for the LM Dictionary — Task

- **Model:** A1 [Solved]
- **Question:** is the weaker same-day LM slope of Lecture 15 (15.5 bp against 39.2 bp for FinBERT) a weaker market reaction, or a worse measurement?
- **Data**
  - ▶ Twitter Financial News, validation set:  $N = 2\,388$  headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
  - ▶ Same model and assumption as A1; now  $\hat{s} = 1$  when the LM tone of the headline is negative
- **Tasks**
  - (a) Compute  $\pi$ ,  $p$ ,  $\alpha_0$  and  $\alpha_1$  from the LM matrix.
  - (b) Compute  $\lambda$  and  $1/\lambda$  with the formula of A1.
  - (c) Compute  $\text{Cov}(u, s^*)$ : is the LM error classical?
  - (d) Compare  $\lambda$  with FinBERT (A1)
    - which attenuates more?
    - by what ratio?
- **Report:** the four rates,  $\lambda$ ,  $1/\lambda$ , the ratio

LM confusion matrix, Twitter

| True \ predicted | neg | neu   | pos | Total |
|------------------|-----|-------|-----|-------|
| neg              | 172 | 166   | 9   | 347   |
| neu              | 247 | 1 192 | 127 | 1 566 |
| pos              | 45  | 349   | 81  | 475   |
| Total            | 464 | 1 707 | 217 | 2 388 |

## A6 [Proposed]: Fleiss' $\kappa$ and One Dawid–Skene Step — Task

### Model: A1 [Solved]

#### Questions

- ▶ how much do raters agree beyond chance?
- ▶ what is the most likely true label of a disputed sentence?

#### Inputs

- ▶ Illustrative votes (made up for the exercise): 6 sentences,  $m = 5$  raters each
- ▶  $P_i = \sum_j n_{ij}(n_{ij} - 1)/[m(m - 1)]$ ,  $\bar{P} = \frac{1}{N} \sum_i P_i$ ,  $p_j$  = share of votes for  $j$ ,  
 $\bar{P}_e = \sum_j p_j^2$ ,  $\kappa = (\bar{P} - \bar{P}_e)/(1 - \bar{P}_e)$  [Fleiss \(1971\)](#)

#### Tasks

- Compute every  $P_i$  and  $\bar{P}$ .
- Compute  $p_j$ ,  $\bar{P}_e$  and  $\kappa$ .
- E-step of [Dawid & Skene \(1979\)](#): each rater gives the true class with probability 0.8 and each other class with 0.1, uniform prior; compute  $P(\text{true} = j \mid \text{votes}) \propto \prod_l \pi_{jl}^{n_{lj}}$  for sentences 4 and 5.
- Compare the posteriors with the vote shares and describe the M-step.

#### Report: $\kappa$ and the two posteriors

#### Interpretation: how would you treat the disputed sentences in a test set?

| Sentence | neg | neu | pos |
|----------|-----|-----|-----|
| 1        | 5   | 0   | 0   |
| 2        | 0   | 5   | 0   |
| 3        | 0   | 1   | 4   |
| 4        | 1   | 3   | 1   |
| 5        | 2   | 3   | 0   |
| 6        | 0   | 2   | 3   |

## A4 [Proposed]: Is "No Difference" a Power Problem? — Task

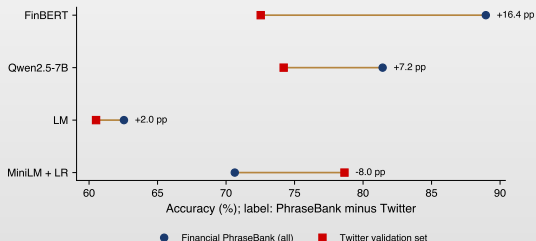
- **Model:** A3 [Solved]
- **Question:** FinBERT and Qwen2.5-7B look tied: is that evidence of equal accuracy, or of a weak test?
- **Inputs**
  - ▶ Twitter validation set,  $n = 2\,388$ ;  $A = \text{FinBERT}$ ,  $B = \text{Qwen2.5-7B (prompt P1)}$ ;  $n_{01} = 395$ ,  $n_{10} = 355$
- **Tasks**
  - (a) Compute  $\hat{d}$ ,  $\hat{q}$ , SE, the 95% CI and  $z$  with the formulas of A3.
  - (b) Run the exact McNemar test at 5%.
  - (c) Compute the power at  $n = 2\,388$  if the true  $d$  is  $\hat{d}$ , and  $n_{80}$ .
  - (d) Use the CI to state which differences the data rule out.
- **Report:** the numbers of (a)–(d)
- **Interpretation:** may you conclude that the two models are equally accurate?

## B3 [Solved]: Training-Data Leakage — Task

- ▣ **Question:** how much of FinBERT's accuracy on Financial PhraseBank comes from having been trained on it?
- ▣ **Data**
  - ▶ Financial PhraseBank [Malo et al. \(2014\)](#): 4 846 sentences (2 264 with full agreement); FinBERT was fine-tuned on a split of them (3 101 of 4 845 in [Araci \(2019\)](#))
  - ▶ Twitter validation set, 2 388 headlines unseen by FinBERT; classifiers: FinBERT, Qwen2.5-7B, LM, MiniLM + logistic regression trained on Twitter [Wang et al. \(2020\)](#)
- ▣ **Tasks**
  - (a) Compute accuracy (2 000-draw bootstrap CI) and macro-F1 on PhraseBank with full agreement, all of PhraseBank and Twitter.
  - (b) Compute the drop from all of PhraseBank to Twitter for each classifier.
  - (c) Compute the difference in differences FinBERT minus Qwen2.5-7B with a bootstrap CI (each data set resampled independently).
- ▣ **Report:** a table of accuracies and drops, the difference in differences with its CI
- ▣ **Interpretation:** how do you separate "harder data" from "memory of the training set"?

## B3 [Solved]: Solution

| Accuracy (%)       | PhraseBank, 100% agr. | PhraseBank, all | Twitter | Drop (pp) | Macro-F1, Twitter |
|--------------------|-----------------------|-----------------|---------|-----------|-------------------|
| FinBERT            | 97.2                  | 89.0            | 72.5    | 16.4      | 66.8              |
| Qwen2.5-7B         | 93.6                  | 81.4            | 74.2    | 7.2       | 72.6              |
| LM                 | 65.0                  | 62.5            | 60.5    | 2.0       | 46.2              |
| MiniLM + log. reg. | 81.4                  | 70.6            | 78.6    | -8.0      | 69.9              |



- (c)  $16.4 - 7.2 = 9.2$  pp from unrounded accuracies, bootstrap CI [6.7; 11.9]: excludes 0
- Qwen2.5-7B was not tuned on PhraseBank labels, so its drop measures the harder data; if both would lose the same without overlap (an untested assumption), FinBERT's extra 9.2 points come from overlap
- Interpretation: MiniLM moves the other way because it is trained on Twitter: every model is best at home

## B4 [Proposed]: How Many Labels Do You Need? — Task

- ▣ **Model:** B3 [Solved]
- ▣ **Question:** how many labelled headlines does a simple supervised model need to beat FinBERT?
- ▣ **Data**
  - ▶ Twitter Financial News: training set of 9 543 headlines, validation set of 2 388; benchmark FinBERT, 72.5%, no Twitter label used
  - ▶ Logistic regression ( $C = 4$ ) on 384-dimensional MiniLM embeddings; logistic regression ( $C = 8$ ) on TF-IDF features of words and word pairs
- ▣ **Tasks**
  - (a) For  $n \in \{100, 200, 400, \dots, 6\,400, 9\,543\}$  draw 10 random training subsets, fit both models and average the validation accuracy.
  - (b) Find the smallest  $n$  at which each model beats FinBERT on average.
  - (c) Report the SD of the embedding accuracy over the 10 repetitions at  $n = 100$ .
- ▣ **Report:** the learning curve and its table
- ▣ **Interpretation:** why do embeddings start higher but grow more slowly?

## B8 [Proposed]: Prompt Sensitivity — Task

- **Model:** B1 [Solved]
- **Question:** does the answer of the same LLM change when the same question is worded differently?
- **Data**
  - ▶ Twitter Financial News, validation set:  $N = 2\,388$  headlines from 2022, each labelled negative, neutral or positive; the human label is treated as the truth
  - ▶ Qwen2.5-7B and 14B with three prompts: P1 "classify the sentiment . . . for investors", P2 "good, bad or neutral for the stock price", P3 "what is the sentiment of this text"
- **Tasks**
  - (a) Compute accuracy with a 2 000-draw bootstrap CI and the share of "neutral" answers for each prompt and model.
  - (b) Run the exact McNemar test of P1 against P3 for each model.
  - (c) Compute the share of headlines with the same label under all three prompts.
- **Report:** a table and a chart of the three prompts
- **Interpretation:** if you picked the best prompt on this set, what would you be doing?

## A5 [Solved]: From Label Probabilities to a Score, and the Optimal Temperature — Task

### ▣ Questions

- ▶ how does the temperature  $T$  change an LLM's probabilities, its score and its calibration?
- ▶ which  $T$  fits the data?

### ▣ Inputs

- ▶ One Twitter headline labelled **neutral**; Qwen2.5-7B (P1) gives  $p_{\text{neg}} = 0.608$ ,  $p_{\text{neu}} = 0.392$ ,  $p_{\text{pos}} = 0.000$  (exactly  $1.0 \times 10^{-12}$ )
- ▶ A random half of the validation set ( $n = 1194$ , seed 15): logits  $z_{ik} = \ln p_{ik}$ , rescaled  $p_{ik}(T) \propto e^{z_{ik}/T}$ , loss  $\text{NLL}(T) = -\sum_i \ln p_{iy_i}(T)$  Guo et al. (2017)

### ▣ Tasks

- Give the label, the net score  $s$  and the logit gap  $\ln(p_{\text{neu}}/p_{\text{neg}})$  of the headline, then its probabilities and  $s$  at  $T = 2$ .
- Explain why labels and accuracy do not depend on  $T$ , while the Brier score does.
- Derive  $\text{NLL}'(T)$ , evaluate the first-order condition at  $T = 1$  and solve for  $T^*$  (Brent on  $[0.5, 50]$ ).
- On the other half, compute ECE and Brier before and after scaling.

### ▣ Report: the label, $s$ , $s(T)$ , $T^*$ , ECE and Brier before and after

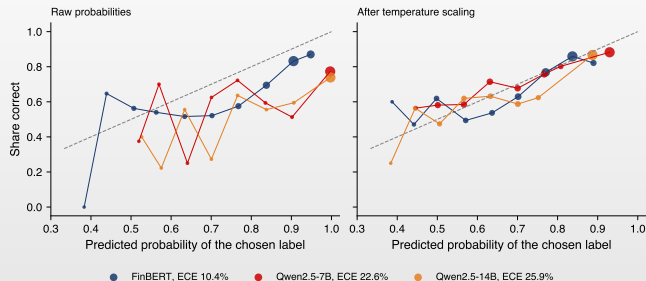
### ▣ Interpretation

- ▶ is the LLM over- or under-confident?
- ▶ what happens to a rule with a threshold on  $s$ ?

## A5 [Solved]: Solution (1/2)

- ▣ (a) Largest probability  $p_{\text{neg}}$ : label negative, wrong;  $s = 0.000 - 0.608 = -0.608$ ; logit gap  $\ln(0.392/0.608) = -0.437$
- ▣ At  $T = 2$ ,  $p_k(T) \propto \sqrt{p_k}$ :  $\sqrt{p} = (0.780, 0.626, 1.0 \times 10^{-6})$ , sum 1.406,  $p(T) = (0.554, 0.446, 0.000)$ ,  $s(T) = -0.554$ : same label, score pulled towards 0
- ▣ (b)  $z \mapsto z/T$  is increasing for  $T > 0$ : the order of the logits, hence  $\arg \max$  and accuracy, do not change; the probabilities, and with them Brier and log loss, do

## A5 [Solved]: Solution (2/2)

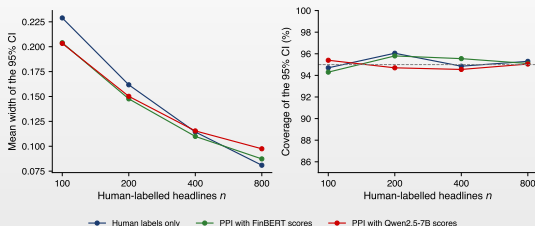


- (c)  $\ln p_{iy}(T) = z_{iy}/T - \ln \sum_k e^{z_{ik}/T}$  gives  $\text{NLL}'(T) = \frac{1}{T^2} \sum_i (z_{iy_i} - \sum_k p_{ik}(T) z_{ik})$
- At  $T = 1$  the average bracket is  $-2.47 < 0$ , so NLL falls as  $T$  grows; the root is  $T^* = 8.43$
- (d) ECE falls from 22.6% to 4.3%, Brier from 0.468 to 0.366 (chart: raw and rescaled reliability)
- Interpretation: raw LLM probabilities are heavily overconfident; a threshold on  $s$  selects different headlines after rescaling

## B9 [Solved]: Prediction-Powered Inference with a Few Human Labels — Task

- **Question:** can a few hundred human labels plus a biased model give a valid and narrower interval for the mean tone of the headlines?
- **Data**
  - ▶ Target  $\theta$  = mean human net tone  $(-1, 0, 1)$  of the 2 388 Twitter validation headlines, which act as the population;  $f$  = net score of FinBERT or of Qwen2.5-7B, fixed
  - ▶ Design:  $n$  labelled headlines (with  $Y$ ) and  $N - n$  unlabelled ones (only  $f$ ), drawn independently with replacement
- **Tasks**
  - (a) For one draw with  $n = 200$ , compute the labels-only estimate  $\bar{Y}_n \pm 1.96 s_Y / \sqrt{n}$ .
  - (b) Compute PPI [Angelopoulos et al. \(2023\)](#):  $\hat{\theta} = \frac{1}{N-n} \sum_{\text{unlab.}} f - \frac{1}{n} \sum_{\text{lab.}} (f - Y)$ ,  $\text{SE} = \sqrt{\hat{V}(f)/(N-n) + \hat{V}(f - Y)/n}$ , for both models.
  - (c) For  $n = 100, 200, 400, 800$ , repeat 2 000 times and record the coverage of  $\theta$  and the mean CI width.
- **Report:** the one-draw estimates and a coverage and width chart
- **Interpretation:** why is the plain model mean  $\frac{1}{N} \sum f$  not an estimate of  $\theta$ ?

## B9 [Solved]: Solution



- True  $\theta = 0.054$ ; plain model means FinBERT  $-0.014$  (wrong sign), Qwen  $+0.101$ : the bias does not shrink as  $N$  grows
- (a) Labels only:  $0.010 \pm 1.96 \times 0.042$ ; (b) PPI-FinBERT: rectifier  $-0.064$ ,  $\hat{\theta} = 0.054 \pm 1.96 \times 0.036$ ; PPI-Qwen: rectifier  $-0.012$ ,  $\hat{\theta} = 0.123 \pm 1.96 \times 0.039$
- (c)  $n = 100$ : coverage 94.7%, 94.3%, 95.4%, width 0.229, 0.204, 0.203 (labels, PPI-FinBERT, PPI-Qwen); gain for large  $N$ : 24% and 27%; at  $n = 800$  PPI is wider (0.087 vs 0.081)
- Interpretation: the model mean converges to the wrong number; the labelled sample corrects it, and PPI is narrower only when  $\text{Var}(f - Y) < \text{Var}(Y)$  and many texts are unlabelled

## B10 [Proposed]: Wild Cluster Bootstrap with 16 Stocks — Task

□ **Model:** B5 [Solved]

□ **Question:** with only 16 stock clusters, which of the nine slopes of B5 remain significant?

□ **Data**

- ▶ FNSPID panel: 18 282 stock-days with news, 16 US stocks, 27.10.2009–18.12.2023;  $z_{i,d}$ : daily net score, standardised over the panel;  $r^e$ : excess return over SPY (%), from the adjusted daily closes of the course data
- ▶ The nine regressions of B5 (three scores  $\times h = 0, 1, 2$ ), SE clustered by stock ( $G = 16$ )

□ **Tasks**

- (a) Compute the  $p$ -value of each slope with  $t_{15}$  critical values.
- (b) Run the wild cluster bootstrap with the null imposed [Cameron, Gelbach & Miller \(2008\)](#):  $y_i^* = \bar{y} + w_{g(i)}(y_i - \bar{y})$ , Rademacher  $w_g = \pm 1$ , 9 999 draws.
- (c) Compute  $t$  with two-way clustering,  $\hat{V} = \hat{V}_{\text{day}} + \hat{V}_{\text{stock}} - \hat{V}_{\text{day} \cap \text{stock}}$ .
- (d) Adjust the nine bootstrap  $p$ -values with Holm at 5%.

□ **Report:** a table with  $t$ , the three  $p$ -values and the two-way  $t$

□ **Interpretation:** does Qwen at  $h = 2$  survive?

## B11 [Proposed]: Specification Curve for the Sentiment Strategy — Task

- ▣ **Model:** B7 [Solved] (a randomisation test)
- ▣ **Question:** over all reasonable variants of the strategy, is there more evidence of predictability than chance would give?
- ▣ **Data**
  - ▶ Sentiment strategy (primer slide 3/3; studied in [Chapter 15](#)): at the close of  $d + 1$  take  $\text{sign}(s_{i,d})$  in every stock with news on day  $d$ , hedged with SPY, closed one day later; four trades on each day with positions; 3 568 trading days
  - ▶ Grid of Lecture 15: three scores  $\times$  thresholds  $|s| > b$ ,  $b \in \{0, 0.25, 0.5\}$ ,  $\times$  return day  $d + 2, \dots, d + 5$ : 36 strategies
- ▣ **Tasks**
  - (a) Compute the mean daily return and  $t_{NW}$  of each strategy and plot the sorted  $t$ -statistics [Simonsohn, Simmons & Nelson \(2020\)](#).
  - (b) Simulate the joint null: flip the signs of the positions in blocks of 10 signal days (dependent wild bootstrap [Shao \(2010\)](#)), the same flips for all 36, 2 000 times.
  - (c) Compute the joint  $p$ -values of the number of  $t > 1.96$  and of the median  $t$ .
- ▣ **Report:** the sorted curve and the two joint  $p$ -values
- ▣ **Interpretation:** what would a "best specification" report hide?

## A8 [Proposed]: The Qwen Strategy after Costs — Task

- ▣ **Model:** A7 [Solved]
- ▣ **Question:** the Qwen2.5-7B strategy has a statistically positive gross mean: is it tradable?
- ▣ **Inputs**
  - ▶ Sentiment strategy (primer slide 3/3; studied in [Chapter 15](#)): at the close of  $d + 1$  take  $\text{sign}(s_{i,d})$  in every stock with news on day  $d$ , hedged with SPY, closed one day later; four trades on each day with positions; 3 568 trading days
  - ▶ Qwen2.5-7B signal:  $\hat{\mu} = 5.64$  bp,  $t_{NW} = 2.26$  ( $L = 8$ ),  $a = 0.950$ , gross Sharpe ratio 0.63
- ▣ **Tasks**
  - (a) Compute the break-even cost  $c^*$ .
  - (b) Compute  $\text{SE}(c^*)$  by the delta method and the 95% CI.
  - (c) Recover  $\sigma$  and compute the net Sharpe ratio at 5 bp.
- ▣ **Report:**  $c^*$ , SE, CI, net Sharpe ratio
- ▣ **Interpretation:** does a significant gross mean imply a profit after costs?

## B6 [Proposed]: Event Study with the Loughran–McDonald Tone — Task

□ **Model:** B5 [Solved]

□ **Questions**

- ▶ around negative and positive news days, when does the price move?
- ▶ does the dictionary separate the two groups as well as FinBERT?

□ **Data**

- ▶ FNSPID panel: 18 282 stock-days with news, 16 US stocks, 27.10.2009–18.12.2023;  $z_{i,d}$ : daily net score, standardised over the panel;  $r^e$ : excess return over SPY (%), from the adjusted daily closes of the course data
- ▶ The daily LM tone is exactly 0 on 48% of stock-days: groups tone  $< 0$  (5 196 stock-days) and tone  $> 0$  (4 288); FinBERT: bottom and top terciles

□ **Tasks**

- For each date, average the excess returns of days  $-5, \dots, +5$  over that date's stock-days in each group.
- Compute the mean over dates of the CAR for  $[-5, -1]$ , day 0, day 1 and  $[+2, +5]$ , with  $t_{NW}$ .
- Plot the mean cumulative paths of the four groups.

□ **Report:** the table of the four windows and the chart

□ **Interpretation:** why is the post-event return positive for both groups [Tetlock \(2007\)](#)?

## C1 [Proposed]: Does Headline Sentiment Pay after Costs? — Task

▣ **Model:** A7, B5 [Solved]

▣ **Question:** does a daily strategy on FNSPID headline sentiment earn money after realistic costs?

▣ **Data**

- ▶ Sentiment strategy (primer slide 3/3; studied in [Chapter 15](#)): at the close of  $d + 1$  take  $\text{sign}(s_{i,d})$  in every stock with news on day  $d$ , hedged with SPY, closed one day later; four trades on each day with positions; 3 568 trading days
- ▶ Positions equally weighted; days without news: return 0; 5 bp per trade; scores FinBERT, Qwen2.5-7B, LM from the saved tables

▣ **Tasks**

- For FinBERT, compute the mean daily gross return,  $t_{NW}$  and a moving-block bootstrap 95% CI (blocks of 10 days, 2 000 draws).
- Compute the Sharpe ratio before and after costs, the break-even cost and the Sharpe ratio at 0, 1, 2, 5, 10 bp.
- Repeat for Qwen2.5-7B and LM, and for strong news  $|s| > 0.5$  (exploratory); Holm over the three unfiltered strategies, or over all six if you select among them.
- Further step: correct the same-day slope with a random human-labelled sample of stock-days (divide by  $\rho(s^*, \hat{s})$ , or rectify the OLS moments as in B9).

▣ **Report:** a two-page note with a results table and charts, and a Quantlet folder

▣ **Interpretation:** discuss timing, look-ahead (B7), survivorship of the 16 stocks and crowding [Lopez-Lira & Tang \(2026\)](#)

## C3 [Proposed]: Replicating Lopez-Lira and Tang on FNSPID — Task

- ▣ **Model:** C1, B5, B10, B7
- ▣ **Question:** does the published GPT-4 design [Lopez-Lira & Tang \(2026\)](#), run with an open model on the course headlines, find the same reaction and the same drift?
- ▣ **Inputs**
  - ▶ Paper: GPT-4 answers YES, NO or UNKNOWN about each headline;  $s = 1, -1, 0$  drives a daily long-short portfolio; overnight drift 0.34% per day, Sharpe ratio 2.97 (Table 1)
  - ▶ Data: FNSPID headlines of the 16 stocks, adjusted daily closes of the course data; first safe return  $d + 2$
  - ▶ Model: Qwen2.5-7B-Instruct [Qwen Team \(2024\)](#), temperature 0, the prompt of Section 5 of the paper word for word
- ▣ **Tasks**
  - (a) Score every headline, parse the first line only, map YES/UNKNOWN/NO to 1/0/−1; report the shares and invalid answers.
  - (b) Average by stock-day and form the daily portfolio with the conventions of the next slide, earning the returns of day  $d + 2$ .
  - (c) Report the mean daily return in bp,  $t_{NW}$ , the hit rate, the Sharpe ratio, and the shares of days with one leg and with no position.
  - (d) Estimate Eq. (1) at headline level for  $h = 0, 2$ , with stock and date fixed effects, SE clustered by date and stock and the wild cluster bootstrap  $p$  (B10).
  - (e) Recompute the Sharpe ratio net of 5, 10, 20 bp per round trip (Figure 4) and compare with C1 on the same stock-days.
- ▣ **Report:** your numbers next to Tables 1 and 2 of the paper, cumulative returns before and after costs, a Quantlet folder
- ▣ **Interpretation**
  - ▶ which differences from the paper can your data separate?
  - ▶ what does B7 say about the bias?

## C3 [Proposed]: Portfolio, Calendar and Cost Conventions

### □ Fixed before any run

- ▶ Weights (Table 1): long leg = equal-weighted mean return of the stocks with  $s > 0$ , short leg = same for  $s < 0$ , a leg only with at least two stocks; long-short = long minus short (gross exposure 2); if only one leg qualifies, hold it alone (footnote 20)
- ▶ Returns: raw stock returns, as in Table 1; the SPY-hedged version (as in C1) is a course adaptation, reported as a sensitivity: with both legs the hedge cancels, so it matters only on one-leg days
- ▶ Calendar: all trading days, as in C1, with return 0 without a position; mean,  $t_{NW}$  and Sharpe ratio over all days; hit rate over days with a position
- ▶ Costs (Figure 4):  $c$  bp per round trip on each unit of notional:  $2c$  per day with two legs,  $c$  with one, plus, in the hedged version,  $c$  on the net SPY notional (1 on one-leg days)
- ▶ Regressions (d): raw returns in % per day, as in Table 2; the date fixed effects absorb the market return; the wild cluster bootstrap keeps both fixed effects in the null model

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